

Учреждение образования
«Брестский государственный технический университет»
Кафедра «ЭВМ и систем»

DESIGN OF INTELLIGENT EDGE AND TIME-SERIES DATA FLOW TEMPERATURE CONTROL SYSTEM

TEXT PROGRAM

(на оптическом носителе CD)

БрГТУ. 41297 - 05 12 00

Листов 4

Объем 2247 КБайт

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Abstract

EdgeHub-Based Closed-Loop Temperature Control System is a software and hardware prototype for automatic temperature stabilization in an edge-control environment. It combines an ESP32-based edge node, MQTT telemetry exchange, a Java Data Hub, and a web HMI into one monitoring and control solution. The edge node reads a digital temperature sensor, filters measurements, executes a PI control algorithm, and drives a PWM heater to keep the process near the configured set point. It also accepts safe parameter updates and reports status without stopping the control process. The Data Hub validates MQTT messages, tracks device state, stores normalized events in TDengine or archive files, and separates raw device traffic from application logic. The HMI exposes REST services and a React interface for live temperature, heater output, control mode, alarms, history, and parameter forms. Operators can review behavior, adjust bounded settings, send commands, and confirm acknowledgements. An AI-assisted tuning workflow analyzes recent telemetry, prepares PID recommendations, checks safety limits, publishes selected values, and compares actual behavior with the predicted response. The program demonstrates the complete path from sensor measurement to actuator response and provides a replaceable basis for future testing and integration with industrial monitoring services.

Content

edge-firmware/ - directory with ESP32 firmware source code;

src/sketch.ino - code file with the firmware entry point;

src/app/edge_app.cpp - code file with edge application workflow;

src/app/runtime_config_store.cpp - code file for runtime parameter storage;

src/controller/pi_controller.cpp - code file with PI temperature control;

src/comms/mqtt/mqtt_gateway.cpp - code file with MQTT connection logic;

src/comms/mqtt/telemetry_builder.cpp - code file for telemetry payloads;

src/comms/mqtt/param_update_handler.cpp - code file for remote parameter updates;

src/hardware/wokwi/ds18b20_sensor.cpp - code file for simulated temperature input;

src/hardware/wokwi/pwm_heater.cpp - code file for simulated PWM heater output;

data-hub/ - directory with Java data-ingestion source code;

src/main/java/com/edgehub/datahub/DataHubApplication.java - code file with service startup;

src/main/java/com/edgehub/datahub/mqtt/ReactiveMqttConsumer.java - code file for MQTT subscription;

src/main/java/com/edgehub/datahub/parser/HubMessageParser.java - code file for topic and payload parsing;

src/main/java/com/edgehub/datahub/pipeline/MqttConsumePipeline.java - code file for message processing;

src/main/java/com/edgehub/datahub/pipeline/DeviceStatusTracker.java - code file for device state tracking;

src/main/java/com/edgehub/datahub/storage/TdengineRestWriter.java - code file for TDengine writes;

hmi/backend/ - directory with FastAPI backend source code;

app/main.py - code file with API application startup;

app/api/routes/devices.py - code file for device API endpoints;

app/api/routes/history.py - code file for telemetry history endpoints;

app/services/mqtt_publisher.py - code file for MQTT command publishing;

app/services/ai/recommendation_service.py - code file for AI recommendation logic;

hmi/frontend/ - directory with React HMI source code;

src/main.tsx - code file with frontend application startup;

src/pages/overview-page.tsx - code file for the monitoring overview screen;

src/pages/device-detail-page.tsx - code file for device details and control;

src/pages/ai-page.tsx - code file for AI tuning and preview;

src/lib/api.ts - code file with the REST API client.

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